

Literature Review: Game Rating Prediction and Recommendation Systems

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Introduction

Predicting video game ratings is a problem with real practical applications - publishers need to estimate market reception before setting marketing budgets, digital storefronts want to highlight quality titles, and indie developers could benefit from data-driven feedback during the design process. The challenge is figuring out what actually makes a game well-received: is it the genre? The platform? The art style? Or something more complex that emerges from how all these features interact? This literature review explores how researchers have tackled rating prediction and recommendation systems, both in the gaming space and in similar domains like movie recommendations.

Recommendation Systems and Review-Based Approaches

Hasan et al. (2024) provide a comprehensive survey of how user reviews have been integrated into recommender systems. Their work is particularly relevant because they emphasize that textual reviews offer much richer information than simple numerical ratings alone. They identify three key advantages of incorporating review text: addressing data sparsity (especially useful when explicit ratings are scarce), improving prediction accuracy through fine-grained details, and enhancing explainability of recommendations.

The authors categorize review-based systems into four main approaches: generic review-based methods that use entire review texts, aspect-based methods that focus on specific features mentioned in reviews (like "graphics" or "storyline" for games), review-rating fusion methods, and ratings-aspect fusion approaches. For our purposes, the aspect-based approach is especially interesting - we can think of game genres, platforms, and other metadata as "aspects" of a game that users care about.

On the technical side, Hasan et al. discuss how deep learning has transformed the field. CNNs can capture local patterns in text (key phrases), RNNs handle sequential information in reviews, and attention mechanisms help models focus on the most important parts of review text. This provides a strong foundation for understanding how neural networks can be applied to rating prediction tasks.

Neural Network Architectures for Rating Prediction

The "Wide & Deep Learning" framework presented by Cheng et al. (2016) offers a particularly influential architecture that's directly applicable to our problem. The key insight is that recommender systems need both memorization (learning specific feature co-occurrences) and generalization (discovering new combinations). Their solution combines a "wide" linear component that handles memorization with a "deep" neural network component that handles generalization.

This architecture is powerful because it jointly trains both components, allowing the model to learn when to rely on simple, memorable patterns and when to generalize. For game rating prediction, this could mean the wide component learns that "puzzle games with pixel art on Switch" tend to score well (a specific pattern it memorizes), while the deep component learns more abstract relationships between game characteristics. The paper also introduces the crucial technique of using embeddings for categorical features - instead of one-hot encoding genres or platforms (which creates huge sparse vectors), embeddings create dense, low-dimensional representations that neural networks can work with much more effectively.

Bayesian Approaches to Rating Prediction

While neural networks excel at finding complex patterns, they can also overfit, especially with sparse data. Lim and Teh (2007) address this problem in the context of movie rating prediction using a Variational Bayesian approach. Their method places prior distributions on model parameters, which helps regularize the model and prevent extreme values that lead to overfitting.

The key advantage of their approach is that instead of finding a single "best" set of parameters, it considers a distribution of possible parameters. This Bayesian treatment provides not just predictions, but also uncertainty estimates - telling us not only what the model predicts, but how confident it is. Their experiments showed that this approach significantly outperformed standard matrix factorization methods and was much more robust to overfitting than maximum a posteriori (MAP) estimation. For predicting ratings of new, unreleased games where we have limited data, this kind of robustness is crucial.

Understanding Game Success Beyond Ratings

Sifa, Drachen, and Bauckhage (2021) provide an interesting counterpoint by analyzing large-scale player behavior data from Steam. Their analysis of over 6 million players and 3,000 games revealed that user review scores show surprisingly weak correlations with actual game ownership and playtime. This finding is important because it suggests that what players say about games (reviews) and what they do (play them) can be quite different.

The authors identified distinct player archetypes and game archetypes based on playtime patterns. Some games show sharp drop-offs in engagement, while others maintain long-term player bases. They also found strong patterns in game ownership - certain game collections tend to co-occur. While their focus is on player behavior rather than rating prediction, their findings suggest that a comprehensive model might need to consider multiple signals beyond just predicting review scores.

Research Gap and Our Approach

The existing literature provides strong foundations for both neural networks and Bayesian approaches to rating prediction. However, most work has focused either on movies (Lim and Teh) or on player behavior without explicit rating modeling (Sifa et al.). Our project aims to apply both an MLP-based approach (informed by the Wide & Deep architecture) and Bayesian modeling specifically to video game rating prediction using game metadata and review text. This allows us to both predict ratings and understand the uncertainty and relative importance of different game features in determining success.

References

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